SHEET 1 — SYSTEM OVERVIEW (§2.1 power path, §2.2 signal path)

PV1 Solar panel
12 V 10 W polycrystalline
Vmp ≈ 18 V, Imp ≈ 0.55 A

PV+ 18 AWG

PV− 18 AWG

BT1 LiFePO4 battery
12.8 V 15 Ah 4S1P
IFR 32140FS, built-in BMS

F1 15 A fuse
≤ 15 cm from BT1 +

BAT+

BAT− 14 AWG

U3 Charge controller
30 A PWM, LCD
mode LI / USER
LOAD = always ON
LVD 12.0 V on LOAD

LOAD− 14 AWG

TB2 GND_BUS
star point (bus bar)
every ground symbol below
returns here

⚠ **SW1 is on the LOAD+ line, not the battery line (§4)**

LOAD+

SW1 ON/OFF
≥ 15 A DC

14 AWG

TB1 +12V_BUS
terminal block

⚠ **NO direct wire from BT1 − to GND_BUS: battery − reaches ground only through U3 (§9)**

IN+ 20–22 AWG

DC+ and COM

+5V → S1–S4 VCC (orange 22 AWG)

S1–S4
HC-SR04 ×4
FL, FR, BL, BR

TRIG GPIO16–19

ECHO
GPIO34/35/36/39
via 1 kΩ / 2 kΩ

Dividers
R4–R7 1 kΩ
R8–R11 2 kΩ

U1 ESP32
DevKit
ESP-WROOM-32
3.3 V logic

VIN

J1 link
unplug for USB

U2 UBEC
12 V → 5 V / 3 A

K1–K3 FC65
relay modules ×3
jumper H
+ R1–R3 10 kΩ at IN

NO→M+

M1–M3 motors
left, right, conveyor
+ D1–D3, C1–C3
at the terminals

GPIO26 / 27 / 25 → K1 / K2 / K3 IN

Wire colour legend (CONNECTIONS.md §1)

- +12 V (BAT+, LOAD+, +12V_BUS, motor +)
- GND / negative (GND_BUS, BAT−, PV−)
- +5 V (UBEC → J1 → VIN, sensor VCC)
- Relay signal (GPIO → FC65 IN)
- Sensor TRIG
- Sensor ECHO

Line weight follows gauge: 14 AWG thickest … 22–24 AWG thinnest.

● Junction (wires joined). No dot = no connection.

⏚ Wire to GND_BUS (TB2, star point, Sheet 2)

PROJECT	**Solar Based Water Boat**
SHEET TITLE	**System overview**
SHEET	**1 of 6**
REV	**A**
DATE	2026-10-24
SIZE / SCALE	A3 landscape · NTS

Source: CONNECTIONS.md
Generated by tools/make_schematic.py — edit the netlist there, then re-run

SHEET 2 — POWER DISTRIBUTION (§3, §4, §5, §7, §9)

PV1 Solar panel 12 V 10 W
poly, Vmp ≈ 18 V, Imp ≈ 0.55 A

18 AWG
+
−
18 AWG

BT1 LiFePO4 12.8 V 15 Ah
4S1P IFR 32140FS, built-in BMS

F1 15 A blade fuse within 15 cm of BT1 +

14 AWG
14 AWG
+
−
14 AWG

PV+
PV−
BAT+
BAT−
LOAD+
LOAD−

U3
30 A PWM solar charge controller (LCD)

Battery type: LI / USER
LOAD mode: always ON (often “24H”)
LVD 12.0 V on LOAD
Settings: Sheet 6

SW1 ON/OFF ≥ 15 A DC

14 AWG
14 AWG

SW1 is on the LOAD+ line, NOT the battery line (§4).

14 AWG

TB1 +12V_BUS terminal block

16–18 AWG → K1 COM · Sheet 3
16–18 AWG → K2 COM · Sheet 3
16–18 AWG → K3 COM · Sheet 3
3 × 20–22 → K1, K2, K3 DC+ · Sheet 3
20–22 AWG

TB2 GND_BUS bus bar — STAR POINT

16–18 AWG → M1 − · Sheet 3
16–18 AWG → M2 − · Sheet 3
16–18 AWG → M3 − · Sheet 3
3 × 20–22 → K1, K2, K3 DC− (+ R1–R3) · Sheet 3
22 AWG → U1 GND · Sheet 4
4 × 22 AWG → S1–S4 GND · Sheet 4
gauge n/s → R8–R11 (2 kΩ) · Sheet 4

U2 UBEC 5 V / 3 A

IN+
IN−
5V OUT
GND OUT

20–22 AWG
22 AWG

J1 5 V link unplug for USB programming (§7)

U1 VIN · Sheet 4
S1–S4 VCC · Sheet 4
+5V · orange 22 AWG

⚠ NO direct wire from BT1 − to GND_BUS (TB2)
Battery − reaches the common ground only THROUGH the controller (BAT− → LOAD−). Many PWM controllers switch the negative side internally; an extra wire bypasses the low-voltage disconnect and can damage U3 (§9).
Star layout: heavy motor returns go straight to TB2, not daisy-chained through the ESP32 ground.

Connection order (§3.2)
1. Connect the battery to BAT+ / BAT− FIRST (F1 fitted, SW1 OFF). U3 powers up and detects 12 V.
2. THEN connect the solar panel to PV+ / PV− (panel covered or face-down).
3. Then connect the loads to LOAD+ / LOAD−.
4. Disconnect in REVERSE: loads, then panel, then battery. Never leave the panel on U3 without the battery.

Notes
All loads (relays, motors, UBEC) connect to the LOAD terminals only, never directly to the battery (§3.1).
Measure U2 output before connecting the ESP32: 4.9–5.2 V (§7).
USB programming: SW1 OFF → unplug J1 → upload → unplug USB → refit J1 → SW1 ON (§7).
Optional battery isolator (§4): a 2nd switch next to F1; open it only after unplugging the panel. Not drawn.

Wire colour legend (CONNECTIONS.md §1)
- +12 V (BAT+, LOAD+, +12V_BUS, motor +)
- GND / negative (GND_BUS, BAT−, PV−)
- +5 V (UBEC → J1 → VIN, sensor VCC)
- Relay signal (GPIO → FC65 IN)
- Sensor TRIG
- Sensor ECHO

Line weight follows gauge: 14 AWG thickest … 22–24 AWG thinnest.
● Junction (wires joined). No dot = no connection.
⏚ Wire to GND_BUS (TB2, star point, Sheet 2)

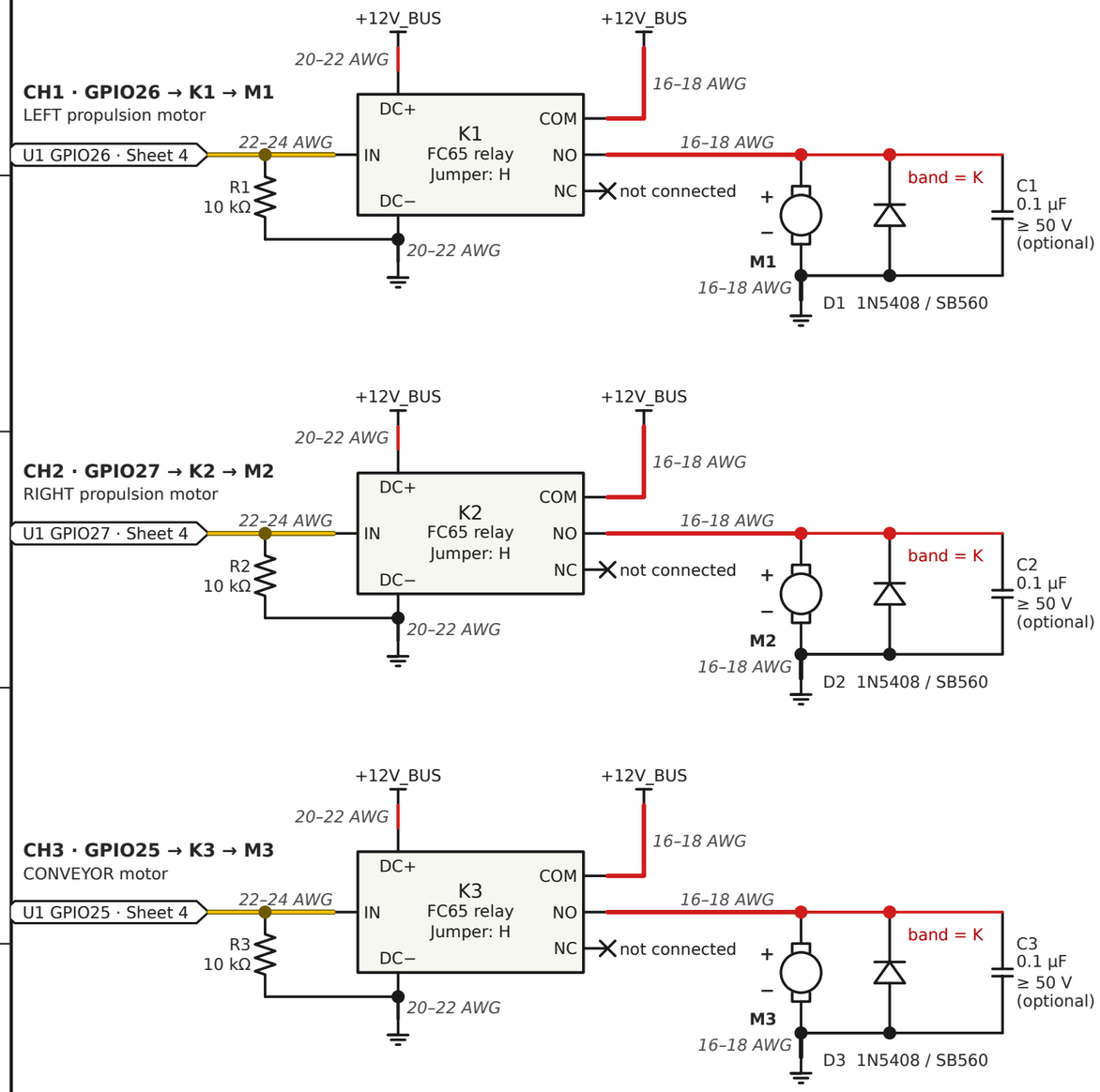
PROJECT
Solar Based Water Boat

SHEET TITLE
Power distribution

SHEET	REV	DATE	SIZE / SCALE
2 of 6	**A**	2026-10-24	A3 landscape · NTS

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SHEET 3 — RELAY AND MOTOR DRIVE (§6.1, §6.2)



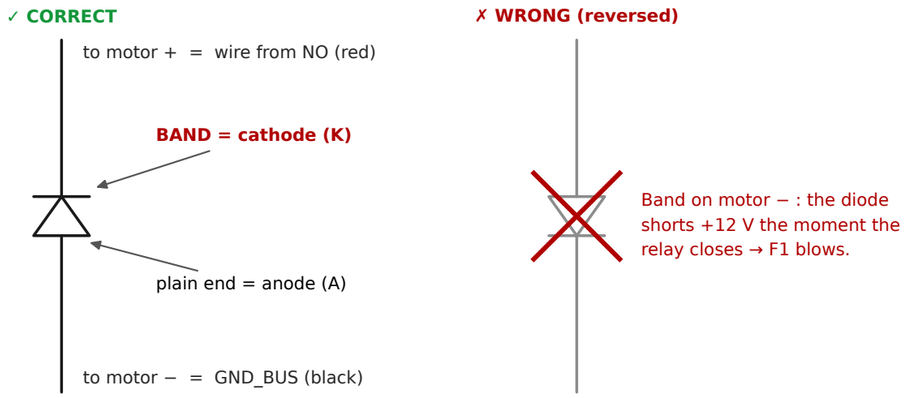
Flyback diode orientation — check every channel (§6.2)

- Band (cathode, K) → motor + = the wire from NO. Plain end (anode, A) → motor − = GND_BUS.
- In normal running the diode is reverse-biased; it only conducts the turn-off spike. Fitted the other way round it shorts 12 V the moment the relay closes and blows F1.
- Solder D1–D3 and C1–C3 directly on the motor terminals, short leads.
- Wrong spin direction? Swap that motor's two wires AT THE MOTOR, then move the diode so the band is still on the wire that comes from NO. The diode follows the wiring, not the "+" marking on the motor.

Relay module notes (§6.1, §6.3)

- Trigger jumper on K1–K3 set to H (HIGH-level): GPIO HIGH = relay ON; GPIO LOW or no signal = OFF. Firmware assumes RELAY_ON_LEVEL HIGH (§6.1).
- R1–R3 (10 kΩ) are soldered across the module's IN and DC− terminals (DC− is on GND_BUS), so IN stays LOW during boot, reset, flashing, or if the yellow wire falls off.
- Use NO only. NC stays unconnected: with no power or signal NO is open and the motor is OFF.
- Before connecting any IN wire to the ESP32: power the module from 12 V, jumper on H, nothing on IN; IN to DC− must read ≈ 0 V. 5–12 V means jumper on L or a pull-up to 12 V — do NOT connect (it would destroy the ESP32 pin).
- Test 3.3 V triggering first (§6.3). If a module does not switch reliably, use the transistor fallback on Sheet 5.

DETAIL — flyback diode orientation at each motor (D1, D2, D3) · illustration, not wiring



Conveyor motor M3 is ASSUMED to be a 12 V DC brushed geared motor — VERIFY before wiring. If it is another voltage or AC it needs its own supply and must not use this circuit (§6.2).

Wire colour legend (CONNECTIONS.md §1)

- +12 V (BAT+, LOAD+, +12V_BUS, motor +)
- GND / negative (GND_BUS, BAT−, PV−)
- +5 V (UBEC → J1 → VIN, sensor VCC)
- Relay signal (GPIO → FC65 IN)
- Sensor TRIG
- Sensor ECHO

Line weight follows gauge: 14 AWG thickest … 22–24 AWG thinnest.

● Junction (wires joined). No dot = no connection.

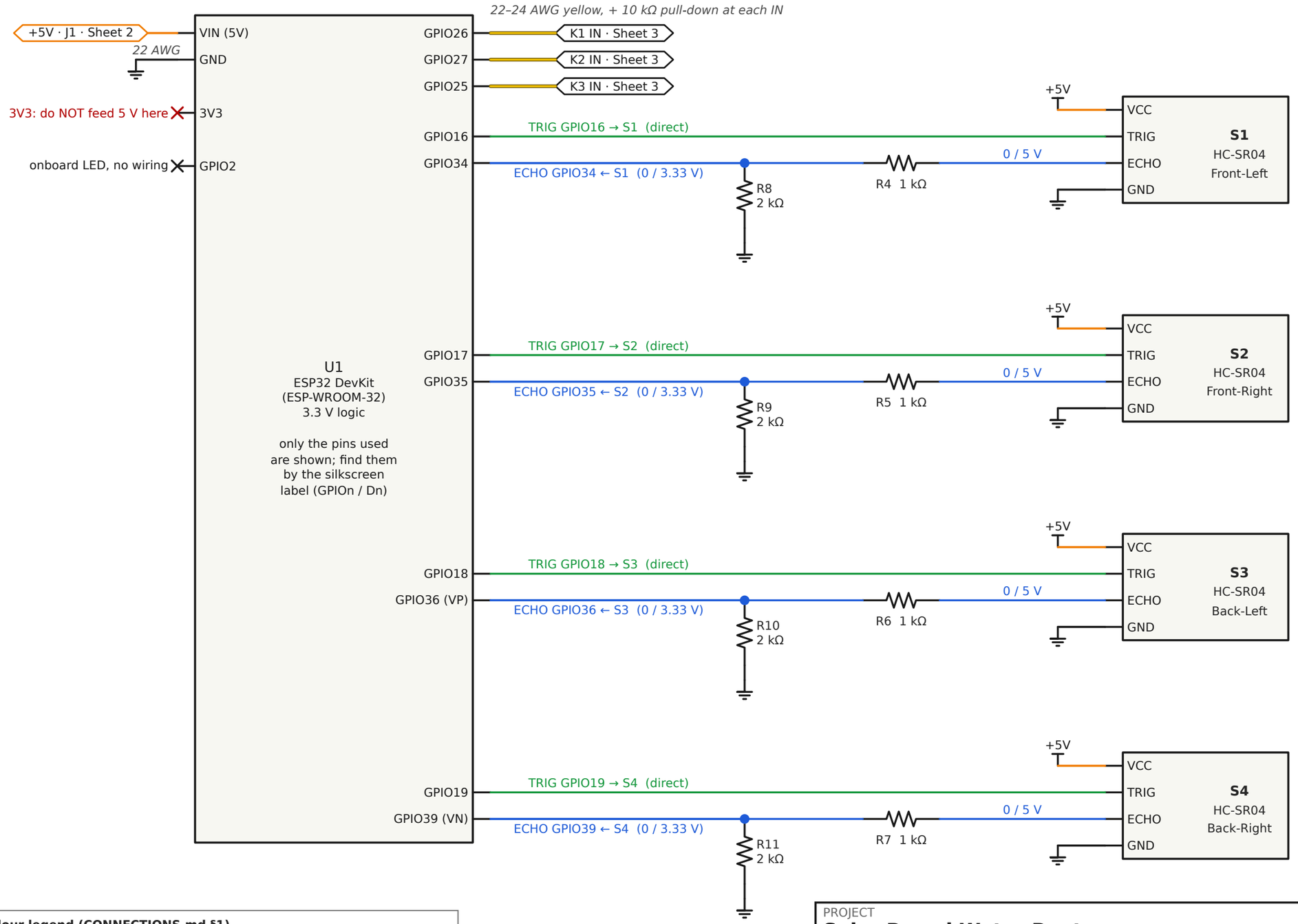
⏚ Wire to GND_BUS (TB2, star point, Sheet 2)

PROJECT	**Solar Based Water Boat**
SHEET TITLE	**Relay and motor drive**
SHEET	**3 of 6**
REV	**A**
DATE	2026-10-24
SIZE / SCALE	A3 landscape · NTS

Source: CONNECTIONS.md

Generated by tools/make_schematic.py — edit the netlist there, then re-run

SHEET 4 — ESP32 AND SENSORS (§2.3, §7, §8)



Notes (§2.3, §8)

- TRIG goes direct: 3.3 V is enough for TRIG.
- ECHO outputs 5 V; ESP32 pins tolerate 3.3 V only — always through the divider.
- GPIO34–39 are input-only and have no internal pull resistors; the 2 kΩ is also their pull-down.
- Solder the dividers on a small perfboard next to the ESP32 (long wires carry 5 V, short ones 3.3 V).
- GPIO25/26/27 stay LOW at boot, so relays cannot click on during start-up.
- VIN may be printed “5V”; GPIO36/39 may be printed VP/VN.
- JSN-SR04T (waterproof) wires the same way, same divider (§10).

ECHO divider math (one per sensor, §8)

HC-SR04 ECHO (0 / 5 V) → 1 kΩ (R4–R7) → node → ESP32 GPIO; node → 2 kΩ (R8–R11) → GND.
Vout = 5 V × 2 kΩ / 3 kΩ = 3.33 V
Use exactly 1 kΩ and 2 kΩ (two 1 kΩ in series make 2 kΩ). 2.2 kΩ gives 3.44 V — above 3.3 V.

Wire colour legend (CONNECTIONS.md §1)

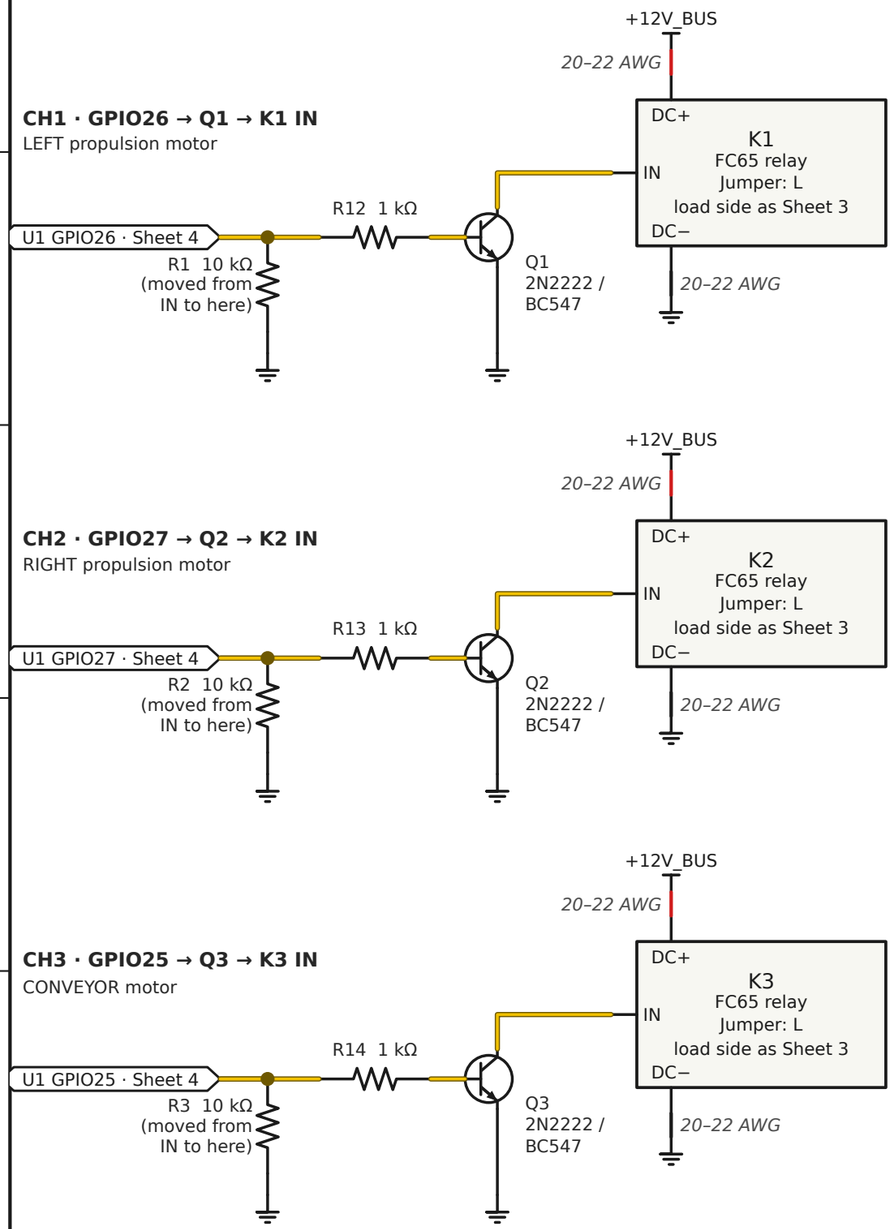
- +12 V (BAT+, LOAD+, +12V_BUS, motor +)
- GND / negative (GND_BUS, BAT−, PV−)
- +5 V (UBEC → J1 → VIN, sensor VCC)
- Relay signal (GPIO → FC65 IN)
- Sensor TRIG
- Sensor ECHO

Line weight follows gauge: 14 AWG thickest … 22–24 AWG thinnest.
● Junction (wires joined). No dot = no connection.
⏚ Wire to GND_BUS (TB2, star point, Sheet 2)

PROJECT	**Solar Based Water Boat**
SHEET TITLE	**ESP32 and ultrasonic sensors**
SHEET	**4 of 6**
REV	**A**
DATE	2026-10-24
SIZE / SCALE	A3 landscape · NTS

Source: CONNECTIONS.md
Generated by tools/make_schematic.py — edit the netlist there, then re-run

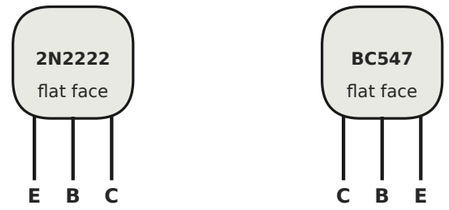
SHEET 5 — TRANSISTOR FALLBACK (OPTIONAL) (§6.3, §6.4)



When and how to use the fallback (§6.3, §6.4)
Use only for a module that fails the 3.3 V trigger test of §6.3 (does not click every time, chatters, or IN current < 0.5 mA). Each channel is converted on its own.

1. Set THIS module's trigger jumper to L (LOW-level trigger).
2. Remove the direct GPIO → IN wire and move the 10 kΩ (R1/R2/R3) from IN to the GPIO side as drawn. Interpretation: §6.4 does not say this explicitly, but with the jumper on L a 10 kΩ left from IN to GND can hold the relay ON.
3. Logic is unchanged, the firmware needs no change: GPIO HIGH → transistor ON → IN pulled LOW → relay ON. No GPIO signal → transistor OFF → IN floats → relay OFF.
4. The ESP32 pin only sees the 1 kΩ base resistor, never 12 V.

Transistor pinouts (§6.4)
2N2222 (TO-92, flat side facing you, legs down): E B C
BC547 (same view): C B E
Pinouts differ between parts and makers — check your datasheet.



TO-92, viewed from the flat side, legs down

Wire colour legend (CONNECTIONS.md §1)
- +12 V (BAT+, LOAD+, +12V_BUS, motor +)
- GND / negative (GND_BUS, BAT−, PV−)
- +5 V (UBEC → J1 → VIN, sensor VCC)
- Relay signal (GPIO → FC65 IN)
- Sensor TRIG
- Sensor ECHO

Line weight follows gauge: 14 AWG thickest … 22–24 AWG thinnest.
● Junction (wires joined). No dot = no connection.
⏚ Wire to GND_BUS (TB2, star point, Sheet 2)

PROJECT
Solar Based Water Boat

SHEET TITLE
Transistor fallback detail

SHEET	REV	DATE	SIZE / SCALE
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Source: CONNECTIONS.md
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SHEET 6 — REFERENCE (§1, §2.3, §3.3, safety notes)

Bill of materials (§1) — designators match Sheets 1–5

Ref	Qty	Part	Value / rating	Source
PV1	1	Solar panel, polycrystalline	12 V 10 W; Vmp ≈ 18 V, Imp ≈ 0.55 A	§1
U3	1	LCD PWM solar charge controller	30 A; must have LI / USER mode	§1, §3
BT1	1	Pro-Range IFR 32140FS LiFePO4 pack	12.8 V 15 Ah 4S1P; check built-in BMS/PCM	§1, §5
F1	1	Blade fuse + inline fuse holder	15 A; on battery +, ≤ 15 cm from battery	§1, §4
SW1	1	Main ON/OFF switch	≥ 15 A DC; on the LOAD+ line	§1, §4
TB1	1	+12 V bus terminal block	Named in §4; not a separate line in §1	§4
TB2	1	Ground bus bar / terminal block	One common ground, star point	§1, §9
U2	1	UBEC	5 V / 3 A, 12 V in	§1, §7
J1	1	2-pin connector / jumper in UBEC 5 V out	Unplug for USB programming	§1, §7
U1	1	ESP32 DevKit	ESP-WROOM-32	§1, §2.3
K1–K3	3	FC65 optocoupler relay module	12 V coil, 30 A contacts; jumper H	§1, §6
M1, M2	2	Rhino IG32 planetary geared motor	12 V 100 RPM; left / right propulsion	§1, §6
M3	1	Conveyor motor	ASSUMED 12 V DC brushed geared — verify	§1, §6.2
D1–D3	3	Flyback diode	1N5408, or Schottky SB560 / SR560 / MBR1045	§1, §6.2
C1–C3	3	Ceramic capacitor (optional)	0.1 µF (100 nF), ≥ 50 V	§1, §6.2
R1–R3	3	Resistor, relay IN pull-down	10 kΩ	§1, §6.1
R4–R7	4	Resistor, ECHO divider (top)	1 kΩ	§1, §8
R8–R11	4	Resistor, ECHO divider (bottom)	2 kΩ — not 2.2 kΩ	§1, §8
S1–S4	4	HC-SR04 ultrasonic sensor	5 V; or JSN-SR04T waterproof	§1, §8, §10
Q1–Q3	3	NPN transistor (fallback only)	2N2222 or BC547	§1, §6.4
R12–R14	3	Resistor, base (fallback only)	1 kΩ	§1, §6.4

Wire colours and gauges (§1)

Colour	Wire	Gauge
Red / Black	Battery & main 12 V line: battery → fuse → controller, LOAD → switch → 12 V bus	14 AWG (2.5 mm²)
Red / Black	Solar panel → controller	18 AWG (0.75–1.0 mm²)
Red / Black	12 V bus → relay COM, relay NO → motor, motor → GND	16–18 AWG (1.0–1.5 mm²)
Red / Black	12 V bus → relay module DC+/DC−, UBEC input	20–22 AWG
Orange	5 V (UBEC → ESP32, sensors)	22 AWG
Yellow	Relay signals (GPIO → IN)	22–24 AWG
Green	Sensor TRIG	22–24 AWG
Blue	Sensor ECHO	22–24 AWG
Black	Logic ground (ESP32, sensors, relay modules)	22 AWG

Final pin map (§2.3)

Function	ESP32 pin	Goes to
Left motor relay	GPIO 26	FC65 #1 (K1) IN + 10 kΩ (R1) to GND
Right motor relay	GPIO 27	FC65 #2 (K2) IN + 10 kΩ (R2) to GND
Conveyor relay	GPIO 25	FC65 #3 (K3) IN + 10 kΩ (R3) to GND
Front-Left TRIG / ECHO	GPIO 16 / GPIO 34	HC-SR04 #1 (S1); ECHO via R4 / R8
Front-Right TRIG / ECHO	GPIO 17 / GPIO 35	HC-SR04 #2 (S2); ECHO via R5 / R9
Back-Left TRIG / ECHO	GPIO 18 / GPIO 36 (VP)	HC-SR04 #3 (S3); ECHO via R6 / R10
Back-Right TRIG / ECHO	GPIO 19 / GPIO 39 (VN)	HC-SR04 #4 (S4); ECHO via R7 / R11
Status LED	GPIO 2	Onboard LED (no wiring)
5 V in	VIN (sometimes 5V)	UBEC 5 V out via J1
Ground	GND	Ground bus (TB2)

U3 charge controller settings for LiFePO4 (§3.3)

Setting	Value
Battery type	LI / USER (never SLD, GEL or FLD)
System voltage	12 V
Charge (boost/bulk) voltage	14.4 V
Float voltage	13.6 V
Equalisation	OFF (or = 14.4 V if it cannot be disabled)
Low-voltage disconnect (LVD)	12.0 V
Low-voltage reconnect (LVR)	12.8 V
Temperature compensation	OFF / 0 mV/°C
LOAD output mode	Always ON (often “24H” or “Normal”); confirm LOAD+ to LOAD− reads battery voltage in daylight

Key safety notes

1. Fit F1 (15 A) FIRST, within 15 cm of the battery +. Keep the battery disconnected until the end.
2. First power-up with the motors DISCONNECTED (README “Bench test”). Check polarity: red +, black −.
3. Before connecting any FC65 IN to the ESP32: module on 12 V, jumper on H, nothing on IN — IN to DC− must read ≈ 0 V. 5–12 V would destroy the ESP32 pin.
4. Never feed 5 V into the ESP32 3V3 pin; 5 V goes to VIN only. Check U2 output is 4.9–5.2 V first.
5. USB programming: SW1 OFF → unplug the J1 5 V link → plug USB and upload → unplug USB → refit J1 → SW1 ON. For Serial debugging with motors powered keep J1 unplugged and run the ESP32 from USB.
6. No wire from battery − to the GND bus; all loads on LOAD+/LOAD− only. Connect battery → panel → loads; disconnect in reverse.
7. Never use relay NC. Diode band on motor + / NO wire — reversed, it shorts 12 V and blows F1.
8. Switch SW1 OFF when the boat is not in use: idle drain (~3 Ah/day) exceeds the panel's ~2.5 Ah/day.

PROJECT
Solar Based Water Boat

SHEET TITLE
Reference: pin map, wires, BOM, settings

SHEET	REV	DATE	SIZE / SCALE
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